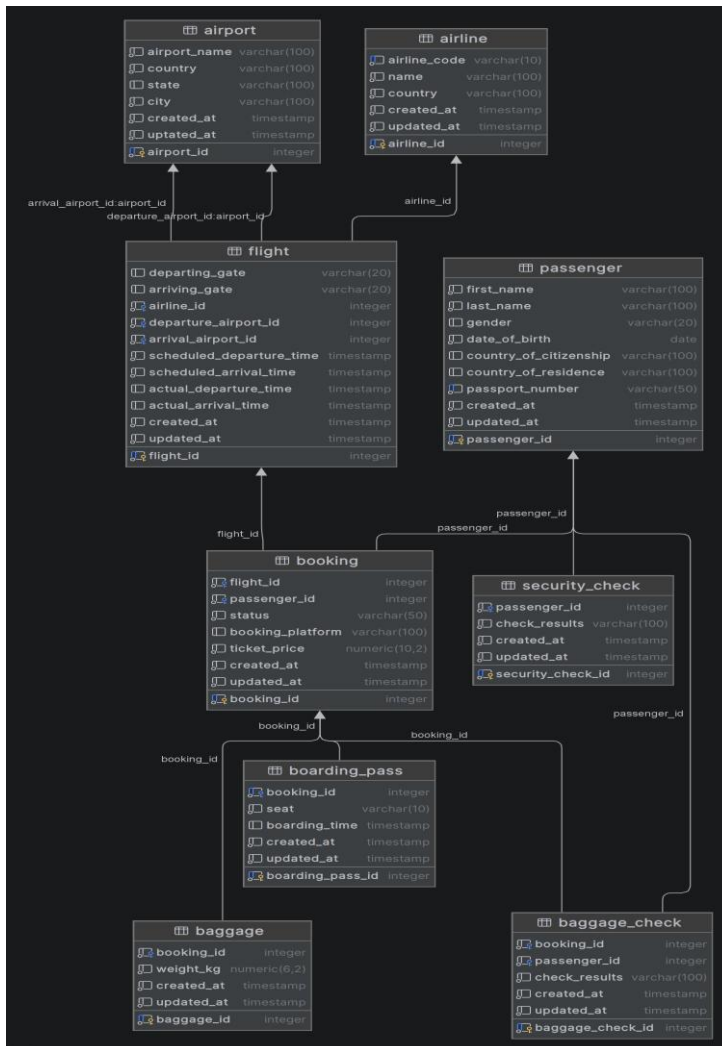


Laboratory Work 1

ERD Diagram



System Description

The International Airport Database is designed to manage information about airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks, and security checks.

The system connects flights with airlines and airports, bookings with passengers and flights, and stores additional information related to baggage and airport security procedures.

Entities and Attributes

1. Airport

Stores information about airports.

Primary Key:

airport_id

Attributes:

airport_id, airport_name, country, state, city, created_at, updated_at

2. Airline

Stores information about airline companies.

Primary Key:

airline_id

Unique Attribute:

airline_code

Attributes:

airline_id, airline_code, name, country, created_at, updated_at

3. Flight

Stores scheduled and actual flight information.

Primary Key:

flight_id

Foreign Keys:

airline_id → Airline

departure_airport_id → Airport

arrival_airport_id → Airport

Attributes:

*flight_id, departing_gate, arriving_gate, airline_id,
departure_airport_id, arrival_airport_id,
scheduled_departure_time, scheduled_arrival_time,
actual_departure_time, actual_arrival_time,
created_at, updated_at*

4. Passenger

Stores passenger information.

Primary Key:

passenger_id

Unique Attribute:

passport_number

Attributes:

*passenger_id, first_name, last_name, gender,
date_of_birth, country_of_citizenship,
country_of_residence, passport_number,
created_at, updated_at*

5. Booking

Stores information about passenger bookings.

Primary Key:

booking_id

Foreign Keys:

flight_id → Flight

passenger_id → Passenger

Attributes:

*booking_id, flight_id, passenger_id,
status, booking_platform, ticket_price,
created_at, updated_at*

6. Boarding Pass

Stores boarding pass information.

Primary Key:

boarding_pass_id

Foreign Key:

booking_id → Booking

Attributes:

*boarding_pass_id, booking_id, seat,
boarding_time, created_at, updated_at*

7. Baggage

Stores baggage information related to bookings.

Primary Key:

baggage_id

Foreign Key:

booking_id → Booking

Attributes:

*baggage_id, booking_id, weight_kg,
created_at, updated_at*

8. Security Check

Stores passenger security check results.

Primary Key:

security_check_id

Foreign Key:

passenger_id → Passenger

Attributes:

*security_check_id, passenger_id,
check_results, created_at, updated_at*

9. Baggage Check

Stores baggage check information.

Primary Key:

baggage_check_id

Foreign Keys:

booking_id → Booking

passenger_id → Passenger

Attributes:

*baggage_check_id, booking_id, passenger_id,
check_results, created_at, updated_at*

Relationships and Cardinalities

1. Airline — Flight

One airline can operate many flights.

Each flight belongs to one airline.

Cardinality:

Airline 1 : N Flight

2. Airport — Flight

One airport can be used as the departure airport for many flights.

Each flight has one departure airport.

Cardinality:

Airport 1 : N Flight

One airport can also be used as the arrival airport for many flights.

Each flight has one arrival airport.

Cardinality:

Airport 1 : N Flight

3. Flight — Booking

One flight can have many bookings.

Each booking belongs to one flight.

Cardinality:

Flight 1 : N Booking

4. Passenger — Booking

One passenger can have many bookings.

Each booking belongs to one passenger.

Cardinality:

Passenger 1 : N Booking

5. Booking — Boarding Pass

One booking can have one boarding pass.

Each boarding pass belongs to one booking.

Cardinality:

Booking 1 : 1 Boarding Pass

6. Booking — Baggage

One booking can have zero or many baggage records.

Each baggage record belongs to one booking.

Cardinality:

Booking 1 : N Baggage

Participation:

Baggage is optional because a passenger may travel without baggage.

7. Passenger — Security Check

One passenger can have many security check records.

Each security check belongs to one passenger.

Cardinality:

Passenger 1 : N Security Check

8. Booking — Baggage Check

One booking can have many baggage check records.

Each baggage check belongs to one booking.

Cardinality:

Booking 1 : N Baggage Check

9. Passenger — Baggage Check

One passenger can have many baggage check records.

Each baggage check belongs to one passenger.

Cardinality:

Passenger 1 : N Baggage Check

Legend

PK — Primary Key

FK — Foreign Key

1 : N — One-to-Many relationship

1 : 1 — One-to-One relationship